Т11

$H_0: \xi \sim p_0(x) = 1\{(0;1)\}$

$H_1: \xi \sim p_1(x) = \frac{e^{1-x}}{e-1}\{(0;1)\}$

а) $n=1$ с уровнем значим. α.

$l(x) = \frac{L_1}{L_0} = \frac{p_1(x)}{p_0(x)} = \frac{e^{1-x}}{e-1} \geq C$; $\underline{G: x \leq A.}$

$P(x \leq A | H_0) = \alpha$

$\int_0^A dx = \alpha \Rightarrow A = \alpha \Rightarrow G: x \leq \alpha$

$W = P(x \leq A | H_1) = \int_0^A \frac{e^{1-x}}{e-1} dx = \frac{e}{e-1} \cdot \frac{e^{-x}}{(-1)} \Big|_0^A = \frac{e}{e-1}(1-e^{-\alpha})$

$\alpha_1 = \alpha$

$\alpha_2 = 1 - W$

б) $\underline{\underline{n=2}}$

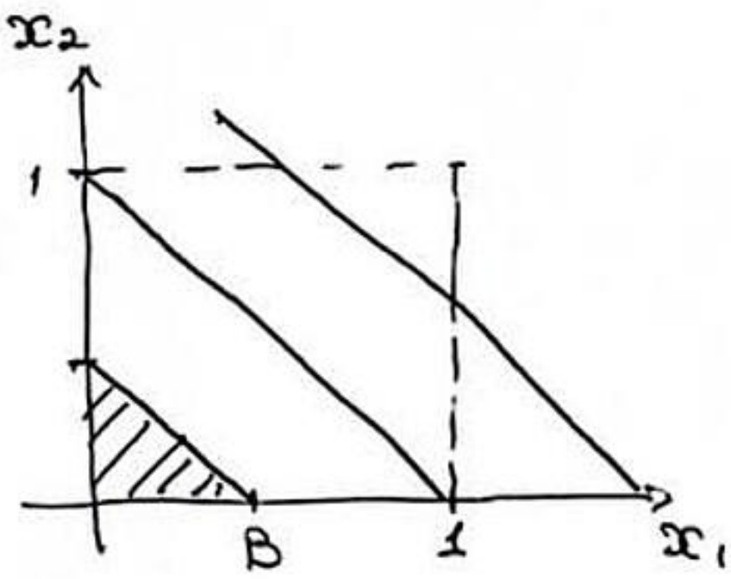
$l = \frac{L_1}{L_0} = \frac{\frac{e^{1-x_1}}{e-1} \cdot \frac{e^{1-x_2}}{e-1}}{1 \cdot 1} = \frac{e^{2-x_1-x_2}}{(e-1)^2} \geq C$

$G: x_1 + x_2 \leq B$

$P(x_1 + x_2 \leq B | H_0) = \alpha$

$\iint_{x_1+x_2 \leq B} 1 \cdot 1 \cdot dx_1 \cdot dx_2 = \alpha$

$\left.\begin{array}{l} x_1+x_2 \leq B \\ 0 \leq x_1 \leq 1 \\ 0 \leq x_2 \leq 1 \end{array}\right\}$ $\frac{B^2}{2} = \alpha \Rightarrow B = \sqrt{2\alpha}$ $(B \leq 1)$



$\sqrt{2\alpha} < 1 \Rightarrow \alpha \leq 1/2$

$G: x_1 + x_2 \leq \underline{B = \sqrt{2\alpha}}$

$W = P(x_1 + x_2 \leq B | H_1) = \iint_{\substack{x_1+x_2 \leq B \\ 0 \leq x_i < 1}} \frac{e^{1-x_1} \cdot e^{1-x_2}}{(e-1)^2} dx_1 dx_2 = \frac{e^2}{(e-1)^2} \int_0^B dx_1 \int_0^{B-x_1} e^{-x_1} \cdot e^{-x_2} dx_2 =$

$= \frac{e}{(e-1)^2} \int_0^B e^{-x_1}(1 - e^{-B+x_1}) = \frac{e^2}{(e-1)^2}[1 - e^{-B} - e^{-B} \cdot B]$, где $B = \sqrt{2\alpha}$

$\begin{cases} \alpha_1 = \alpha \\ \alpha_2 = 1 - W \end{cases}$

в) Асимптот. критерий по выборке n

$l(\overline{x_n}) = \frac{L_1(\overline{x_n})}{L_0(\overline{x_n})} = \frac{\prod_{i=1}^n p_1(x_i)}{\prod_{i=1}^n p_0(x_i)} = \prod_{i=1}^n \frac{p_1(x_i)}{p_0(x_i)} \geq C$

$\ln l = \sum_{i=1}^n \ln \frac{p_1(x_i)}{p_0(x_i)} \geq \ln C \Rightarrow \sum_{i=1}^n \ln p_1(x_i) - \sum_{i=1}^n \ln p_0(x_i) = \sum_{i=1}^n \ln \frac{e^{1-x}}{e-1} - 0 \geq \ln C$

$$\sum_{i=1}^{n} \ln(1-x_i) - n\ln(e-1) \geq \ln C$$

$$G: \sum_{i=1}^{n} x_i \leq A \Big| \cdot \frac{1}{n} \Rightarrow \bar{x} \leq B \quad ; P(\bar{x} \leq B | H_0) = \alpha$$

ЦПТ

$$\frac{\bar{x} - M\xi}{\sqrt{D\xi}} \sqrt{n} \sim N(0;1)$$

$$H_0: M\xi = 1/2 \quad ; P(\bar{x} \leq B | H_0) = P\left(\frac{\bar{x} - 1/2}{\sqrt{1/12}}\sqrt{n} \leq \underbrace{\frac{B - 1/2}{\sqrt{1/12}}}_{u_\alpha}\sqrt{n}\right)$$

$$D\xi = 1/12$$

$$B = \frac{1}{2} + \frac{u_\alpha}{\sqrt{12n}}$$

$$W = P(\bar{x} \leq B | H_1)$$

$$H_1: M\xi = \int_0^1 x p_1(x)\,dx = \frac{e-2}{e-1}$$

$$M\xi^2 = \int_0^1 x^2 p_1(x)\,dx = \frac{2e-5}{e-1}$$

$$D\xi = \frac{e^2 - 3e + 1}{(e-1)^2}$$

$$P\left(\frac{\bar{x} - M\xi}{\sqrt{D\xi}}\sqrt{n} \leq \overbrace{\frac{B - M\xi}{\sqrt{D\xi}}\sqrt{n}}^{\gamma}\right) = \int_{-\infty}^{\gamma} \frac{1}{\sqrt{2\pi}} e^{-x^2/2}\,dx =$$

$$= \frac{\frac{u_\alpha}{\sqrt{12}} + 0{,}082\sqrt{n}}{\sqrt{D\xi}} \xrightarrow[n\to\infty]{} +\infty$$

$$\Rightarrow W \xrightarrow[n\to\infty]{} 1 \quad \left(\text{Интеграл } \int \frac{1}{\sqrt{2\pi}} e^{-x^2/2}\,dx \xrightarrow[n\to\infty]{} 1\right)$$

$$\begin{cases} \alpha_1 = \alpha \\ \alpha_2 = 1 - W \end{cases}$$

2) Построить оптим. критерий по выборке объёма n с крит. обл G, $x_{min} < C$ и ур-ем значимости α.

$$\alpha_2 \to \min \qquad \alpha_1 = P(x_{min} \leq C | H_0) = 1 - (1 - F(C))^n \leq \alpha$$

$$\alpha_1 \leq \alpha$$

$$1 - (1 - C)^n \leq \alpha \Rightarrow 1 - \sqrt[n]{1-\alpha} \geq C$$

$$\alpha_2 = P(x_{min} > C | H_1) = 1 - P(x_{min} \leq C | H_1) = (1 - \Phi(C))^n) =$$

$$\left\{\Phi(C) = \frac{e}{e-1}(1 - e^{-C})\right\} = \left(1 - \frac{e}{e-1}(1 - e^{-C})\right)^n$$

$$\frac{d\alpha_2}{dc} = n\left(1-\frac{e}{e-1}(1-e^{-c})\right)^{n-1}\cdot\left(-\frac{e^{1-c}}{e-1}\right) = 0$$

$\alpha_2 \to \min$
$\alpha_1 \le \alpha$ $\longrightarrow$ $1-\sqrt[n]{1-\alpha} \geq c$

$$\frac{e}{e-1}(1-e^{-c}) = 1$$

т.к. $1-\alpha \in (0;1) \Rightarrow \sqrt[n]{1-\alpha} \in (0;1) \Rightarrow$ $\quad e^{-c} = e^{-1} \Rightarrow \boxed{c' = 1}$

$\Rightarrow 1-\sqrt[n]{1-\alpha} \in (0;1) \Rightarrow c' > 1-\sqrt[n]{1-\alpha} \Rightarrow$ усл. для α_2 ✓

$$\frac{d^2\alpha_2}{dc^2} = n(n-1)\left(1-\frac{e}{e-1}(1-e^{-c})\right)^{n-2}\left(-\frac{e^{1-c}}{e-1}\right)^2 + n\left(1-\frac{e}{e-1}(1-e^{-c})\right)^{n-1}\left(\frac{e^{1-c}}{e-1}\right) > 0$$

$\Rightarrow c$ – точка минимум. $\Rightarrow$ т.к. $c \leq 1-\sqrt[n]{1-\alpha}$, то $c = 1-\sqrt[n]{1-\alpha}$. $\Rightarrow$
$(\alpha_1 = \alpha)$

$\Rightarrow C: x_{min} < 1-\sqrt[n]{1-\alpha}$

$$W = P(x_{min} < 1-\sqrt[n]{1-\alpha} \mid H_1) = 1-(1-P(c))^n = 1\left(1-\frac{e}{e-1}(1-e^{-c})\right)^n =$$

$$= 1-\left(1-\frac{e}{e-1}\left(1-e^{-1+\sqrt[n]{1-\alpha}}\right)\right)^n \textcircled{=}$$

$$-1+\sqrt[n]{1-\alpha} = -1+e^{\frac{1}{n}\ln(1-\alpha)} = -1+1+\frac{1}{n}\ln(1-\alpha)+o\left(\frac{1}{n}\right) = \frac{1}{n}\ln(1-\alpha)+o\left(\frac{1}{n}\right)$$

$$e^{-1+\sqrt[n]{1-\alpha}} = e^{\frac{1}{n}\ln(1-\alpha)+o\left(\frac{1}{n}\right)} = 1+\frac{1}{n}\ln(1-\alpha)+o\left(\frac{1}{n}\right)$$

$$\textcircled{=}\ 1-\left(1-\frac{e}{e-1}\left(1-1-\frac{1}{n}\ln(1-\alpha)+o\left(\frac{1}{n}\right)\right)\right)^n = 1-\left(1+\frac{e}{e-1}\frac{1}{n}\ln(1-\alpha)+o\left(\frac{1}{n}\right)\right)^n \longrightarrow$$

$$\xrightarrow{n\to\infty} 1-e^{\frac{e\ln(1-\alpha)}{e-1}} = 1-(1-\alpha)^{\frac{e}{e-1}} \neq 1 \Rightarrow$$ <u>критерий не состоятелен</u>

Т12 H_0 ; $p_4 = \frac{8}{24}$; $p_3 = \frac{4}{24}$; $p_2 = p_1 = \frac{6}{24}$ $\quad \begin{cases} \alpha = 0{,}2 \\ n = 2 \end{cases}$

H_1 : $p_1 = p_2 = p_3 = p_4 = \frac{6}{24}$

$$\ell(\vec{x_n}) = \frac{L_1(\vec{x_n})}{L_0(\vec{x_n})}$$

p_0:

H_0	1	2	3	4
1	1/16	1/16	1/24	1/12
2	1/16	1/16	1/24	1/12
3	1/24	1/24	1/36	1/18
4	1/12	1/12	1/18	1/9

p_1:

H_1	1	2	3	4
1	1/16	1/16	1/16	1/16
2	1/16	1/16	1/16	1/16
3	1/16	1/16	1/16	1/16
4	1/16	1/16	1/16	1/16

ℓ:

ℓ	1	2	3	4
1	1	1	3/2	3/4
2	1	1	3/2	3/4
3	3/2	3/2	9/4	9/8
4	3/4	3/4	9/8	9/16

$p_{ij} = p_i \cdot p_j$ $\qquad$ $P(\ell \geq C \mid H_0) \leq 0{,}2$

Идём по таблице ℓ от max к min и набираем элементы до тех пор, пока $\sum p_{0i} < 0{,}2$

$\cancel{\frac{9}{4} + \frac{3}{2} + \frac{3}{2}}$

$\frac{1}{24} \cdot 4 + \frac{1}{36} = \frac{7}{36} < 0{,}2$ ✓ $\qquad$ $\frac{1}{24} \cdot 4 + \frac{1}{36} + 1/18 = 9/36 = 0{,}25 > 0{,}2$ ⊗

$\Updownarrow \qquad \Updownarrow$

$\ell = \frac{5}{2} \qquad \ell = 9/4$

$\Rightarrow \ell = 3/2$; $G: \ell \geq 3/2$ – крит. обл.

$W = 5 \cdot \frac{1}{16} = \frac{5}{16}$; $\alpha_1 = \frac{7}{36}$; $\alpha_2 = 1 - W = \frac{11}{16}$

T13

$\alpha = 0{,}05$

Елисей

$n = 139$ $\quad \sigma_{дл} = 5{,}722$ мм $\quad \sigma_{шир} = 4{,}612$ мм

Ельрона:

$W - ?$

$m = 1000$ $\quad \sigma_{дл} = 6{,}161$ мм $\quad \sigma_{шир.} = 3{,}055$ мм

$\xi \sim N(\theta_1, \theta_2^2)$ — длина

$\eta \sim N(\theta_1', \theta_2'^2)$ — ширина

Длина: $\quad$ Ширина:

1) $H_0: \theta^2_{2ельр} = \theta^2_{2ел.}$ $\quad$ 2) $H_0: \theta'^2_{2ельр} = \theta'^2_{2ел.}$

$H_1: \theta^2_{2ельр} \neq \theta^2_{2ел.}$ $\quad$ $H_1: \theta'^2_{2ельр} \neq \theta'^2_{2ел}$

$S^2 = \frac{n}{n-1}\sigma^2$ $\quad S^2_{дл.ел} = 32{,}8785$ $\quad S^2_{шир.ел.} = 21{,}4246$

$S^2_{дл.ельр} = 37.8859$ $\quad S^2_{шир.ельр} = 25{,}5786$

$$\tilde{\Delta}_{дл} = \frac{S^2_{дл.ел}}{S^2_{дл.ельр}} = 0{,}86735 \qquad \tilde{\Delta}_{шир} = \frac{S^2_{шир.ел}}{S^2_{шир.ельр}} = 0{,}8376$$

$\tilde{\Delta} \sim F(n-1, m-1) = F(138, 999) \Rightarrow G: \tilde{\Delta} \leq F_{\frac{\alpha}{2}}(138, 999)$ или $\tilde{\Delta} \geq F_{1-\frac{\alpha}{2}}(138, 999)$

$F_{\frac{\alpha}{2}}(138, 999) = 0{,}76738$ $\quad F_{1-\frac{\alpha}{2}}(\ldots) = 1{,}27169$

$\Rightarrow \tilde{\Delta}_{шир} \notin G$ и $\tilde{\Delta}_{дл} \notin G \Rightarrow$ Нет оснований отвергнуть H_0 в обоих случаях.

$$W = P(\Delta \in G \mid H_1) = P(\tilde{\Delta} \leq F_{\frac{\alpha}{2}}(138, 999) \mid H_1) + P(\tilde{\Delta} \geq F_{1-\frac{\alpha}{2}}(138, 999) \mid H_1) =$$

$$= P\left(\frac{S^2_{ел}}{S^2_{ельр}} \cdot \frac{\sigma^2_{ельр}}{\sigma^2_{ел}} \leq F_{\frac{\alpha}{2}}(138, 999)\right) + P\left(\frac{S^2_{ел}}{S^2_{ельр}} \cdot \frac{\sigma^2_{ельр}}{\sigma^2_{ел}} \geq F_{1-\frac{\alpha}{2}}(138, 999)\right) \Rightarrow$$

$$\Rightarrow W = \int_{\frac{F_{\frac{\alpha}{2}}(138, 999)}{\theta}}^{+\infty} q(t)\,dt + \int_{0}^{\frac{F_{1-\frac{\alpha}{2}}(138, 999)}{\theta}} q(t)\,dt =$$

Т14 $\alpha = 0{,}05$ $H_0: a = b$
$H_1: a > b$

$\xi \sim N(a, \sigma_x^2)$ $\sigma_x^2 = 2$ $\vec{x_n} = [-1{,}11; -6{,}10; 2{,}42]$ $n = 3$

$\eta \sim N(b, \sigma_y^2)$ $\sigma_y^2 = 1$ $\vec{y_m} = [-2{,}29; -2{,}31]$ $m = 2$

Фишера $\Rightarrow \dfrac{\bar{x} - a}{\sigma_x}\sqrt{n} \sim N(0;1)$; $\dfrac{\bar{y} - b}{\sigma_y}\sqrt{m} \sim N(0;1)$

$\bar{x} - a \sim N(0, \frac{\sigma_x^2}{n})$
$\bar{y} - b \sim N(0, \frac{\sigma_y^2}{m})$ $\Rightarrow \bar{X} - \bar{Y} - (a - b) \sim N(0, \frac{\sigma_x^2}{n} + \frac{\sigma_y^2}{m})$

$$\Delta = \frac{\bar{X} - \bar{Y} - (a - b)}{\sqrt{\frac{\sigma_x^2}{n} + \frac{\sigma_y^2}{m}}} \sim N(0,1) \; ; \; G: \Delta \geq U_{1-\alpha}$$

При условии H_0: $(a = b)$

$$\Delta = \frac{\bar{X} - \bar{Y}}{\sqrt{\frac{\sigma_x^2}{n} + \frac{\sigma_y^2}{m}}} \sim N(0,1)$$

При условии H_1: $(a > b)$; $a - b = \Theta$

$\bar{x} - \bar{y} \sim N(\Theta, \frac{\sigma_x^2}{n} + \frac{\sigma_y^2}{m})$

$W = P(\tilde{\Delta} \in G | H_1) = P(\tilde{\Delta} \geq U_{1-\alpha}(\Theta, \frac{\sigma_x^2}{n} + \frac{\sigma_y^2}{m}))$